

Proving HC (6) with Kempe-Based Method

HC (6): Proposition that if a simple graph has no K_6 minor it is 5 colourable.

- It is apparently true since it “reduces to 4CT [Robertson, Seymour, Thomas]”.

What we know:

Four Colour Theorem: Any simple planar graph (that has no $K_{3,3}$ or K_5 minor – Wagner 1937) is 4-colourable.

HC (5): A simple graph with no K_5 minor is 4-colourable. This is apparently “equivalent to 4CT”.

How is HC (5) equivalent to the 4CT?

- ➔ 4CT says any graph with no $K_{3,3}$, or K_5 minor is 4 colourable, HC (5) says just no K_5 minor.

This means that a planar graph with no K_5 minor, but with $K_{3,3}$ minors, is 4 colourable.

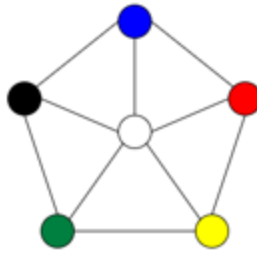
Statement: Any maximally K_5 graph G is 5 colourable. A maximally K_5 graph is a graph G where all vertices are part of a K_5 network. Given a vertex v in the maximally K_5 graph, there exists 4 other vertices such that the 5 vertices form K_5 for all v in G . Every vertex has 4k edges. A maximally K_5 graph has no K_6 minor

Proof: Induction.

Base case: K_5 , K_5 is 5 colourable.

Inductive step: Assume $P(k)$ is true for k vertices, or $k-4$ K_5 graphs, add another node and connecting it to 4 other vertices, then colour the added node the remaining of the 5 colours.

Let's assume a finite graph with no K_6 minors and no $K_{3,3}$ minors, this will be 5 colourable, using Kempe's method with the additional case.



This finite graph G with no K_6 and no $K_{3,3}$ minors has a planar and various maximally K_5 networks within the graph. Both the planar and maximally network will be finite.

Since the maximally K_5 network is finite, there will be a node with only 4 neighbours.

Every finite, maximally K_5 graph is 4-degenerate:

Base case: K_5 , has 5 vertices v and $v-4$ K_5 networks inside. Every vertex has 4 neighbours.

Inductive step: assume $P(k)$ is true for some k vertices or $k-4$ K_5 s. Add a vertex and connect it with 4 vertices to make the graph maximally K_5 , this vertex guarantees the graph to be 4 degenerate. One can add more than 4 edges or else K_6 or make it so that the vertex is not

Since every K_5

This battle with K_5 is not so easy it seems

- Try original method with coloured nodes, may work.
- Give up.

Findings/conjectures today:

- With original K_5 , an added node to the k_5 network can't be connected to > 4 vertices in K_5 network, otherwise a K_6 minor exists. Previously added nodes can have >4 neighbours though.
- The bordering nodes between K_5 network and planar component can't be connected to more than 3 nodes in the K_5 network otherwise it is part of the K_5 network with every node as part of a K_5 minor.

- Proving the arity < 6 or this planar region is ongoing work. It seems that if every node is connected the 3 K_5 network nodes and 3 other nodes in planar network, it is not possible as a $K_{3,3}$ minor exists, but we don't care about $K_{3,3}$, irrelevant.

I think that applying Kempe on a graph with no K_5 and a planar section is 5-colourable since a maximally K_5 is 5-colourable and then applying Kempe on the remaining planar part works with extra case of 5 different colours around a planar node, but no Kempe chain implies toggle. Can Kempe's method be applied.

Deadlock can occur so the original K_5 network colouring can be toggled, can't delete nodes with my strategy.

Question: can a planar graph with K_5 embeddings be used with the recursive method?

Next $K_{3,3}$ embeddings? If both then we prove $hc(6)$.

- Don't consider deadlock, it seems futile. It definitely is for K_5 , maybe not for $K_{3,3}$?
- I think K_5 embedding works since it's a finite graph and K_5 embeddings have a degree of 4 less than 6.
 - o I'm not sure though...
 - o Last question: can Kempe's recursive method work? Is arity < 6 still?
 - o Average seems to be less than 8, but by finiteness, it can be reduced for sure.
 - o SO the answer is yes.
- Think of $K_{3,3}$ now, it's a big thing to gain more insight on, K_5 can be finalized with last question.
 - o Find arity of $K_{3,3}$, check arity of K_5 again...
- $K_{3,3}$'s really not that easy.
- Write something up for what we have so far and the so far coming ideas.

A good idea is to bring my ideas, questions, and approaches and ask them.

Consider $K_{2,2}$, this seems critical in understanding $K_{3,3}$ role in $HC(5)$ vs $4CT$.

- Probably vanity considering $K_{2,2}$ it's just a circuit with 4 nodes.

Moment vs torque, finish 320, 309, 325, 321 edging, actually do the work. WRITE!

APPLY JOBS, write portfolio fun,

Write my ideas down or else they're gonr!

Newcombe's Paradox:

The goal is to get the most money, so you want the hidden box to be chosen or the predictor to think you are going to choose it, then you choose it and since it's right then there'll be money in there. Your decision is your action and it predicts your action, hence decision. So decision must be 1 box.

Lets say you have a graph with maximally k_5 components, planar components and wagner graphs. Remove the minimal amount of edge on K_5 components to make planar, by $Hc(4)$ color it, then

Ideas:

- K_5 , $K_{3,3}$, (Wagner), planar, remove $K + 1$ edges where K is number of added nodes to maximally k_5 network starting from K_5 , to obtain a graph with only planar and $K_{3,3}$, (wagner graphs), by $Hc(4)$, it is 4 colourable, then add edges back and color any repeated colours a fifth colour. This edge removal seems suspicious.
- The degeneracy of K_5 , $K_{3,3}$, and planar is ≤ 5 , implies can apply Kempe's algorithm with one more trivial case.
- Wagner's theorem doesn't include 4-sum since K_4 , can be generated by 1-, 2-, 3-sums, while an K_q can not be generated by $[0, q-1]$ sums for $q = [2, 3]$
- A wagner graph can contract any 2 distinct pairs of nodes, chosen before any initial contraction, so that the $K_{3,3}$ is still apparent, given that the pair of nodes contracted doesn't "cross the centre or one of the centre crossing edges".

- Wait, cases on kempe's method might not work if crossing occurs, ascertain the crossing is not $k_{3,3}$, and just $k_{5,5}$? or something? But that shouldn't matter....?
 - o INVESTIGATE

Investigate the precolouring method.

Investigate the counterexamples as suggested by Prof and document them

Prove the degeneracy relationship.

Understand the Waterloo paper thing, that has some insights

Study prior notes.

Abstract:

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Degeneracy(